

Technical Note: Acoustic and Physical Characterization of Human Breast Skin Layers for Ultrasound Simulation

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1 Introduction

This report consolidates the acoustic and physical parameters for human breast skin, specifically the epidermis and dermis layers, to facilitate high-fidelity ultrasound simulations. For the purposes of computational modeling, the skin is treated as a multi-layered medium. While biological skin includes the hypodermis (subcutaneous fat), this layer has been integrated into the bulk breast substance in our current simulation parameters to streamline the acoustic impedance transitions.

1.1 Anatomical Considerations

According to *Rook's Textbook of Dermatology* (Griffiths et al., 2024), the skin is a complex organ comprising the superficial, non-vascularized **epidermis** and the deeper, fibrous **dermis** (Figure 1). The epidermis acts as the primary barrier, characterized by keratinocytes, while the dermis provides structural integrity through a dense network of collagen and elastin fibers. This histological distinction is critical as the varying protein concentrations directly influence the Speed of Sound (SoS) and density gradients within an ultrasonic field.

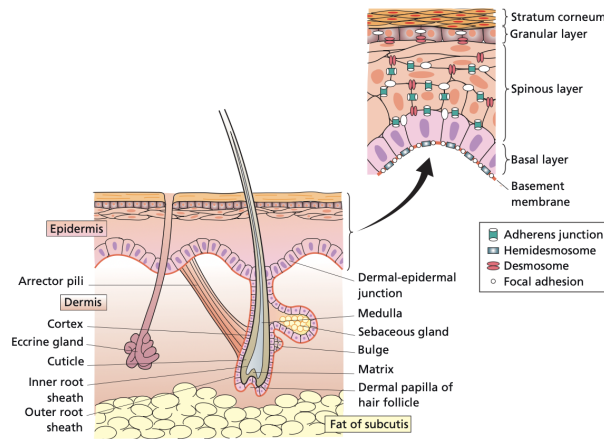


Figure 1: Schematic representation of the skin layers and appendages. In our model, the focus is localized to the Epidermis and Dermis, with the Fat of subcutis (hypodermis) being assimilated into the underlying substance. Adapted from Rook's Textbook of Dermatology (10th Ed.).

1.2 Computational Phantom Generation

To validate beamforming algorithms and phase aberration correction techniques, we developed a series of numerical phantoms. These "Breast Medium Maps" (Figure 2) represent the spatial distribution of SoS and Density. By assigning the calculated acoustic properties (derived in Section 2) to specific grid regions, we can simulate the propagation of ultrasonic waves through realistic biological geometries, including simulated epidermal/dermal boundaries and internal inclusions like tumors.

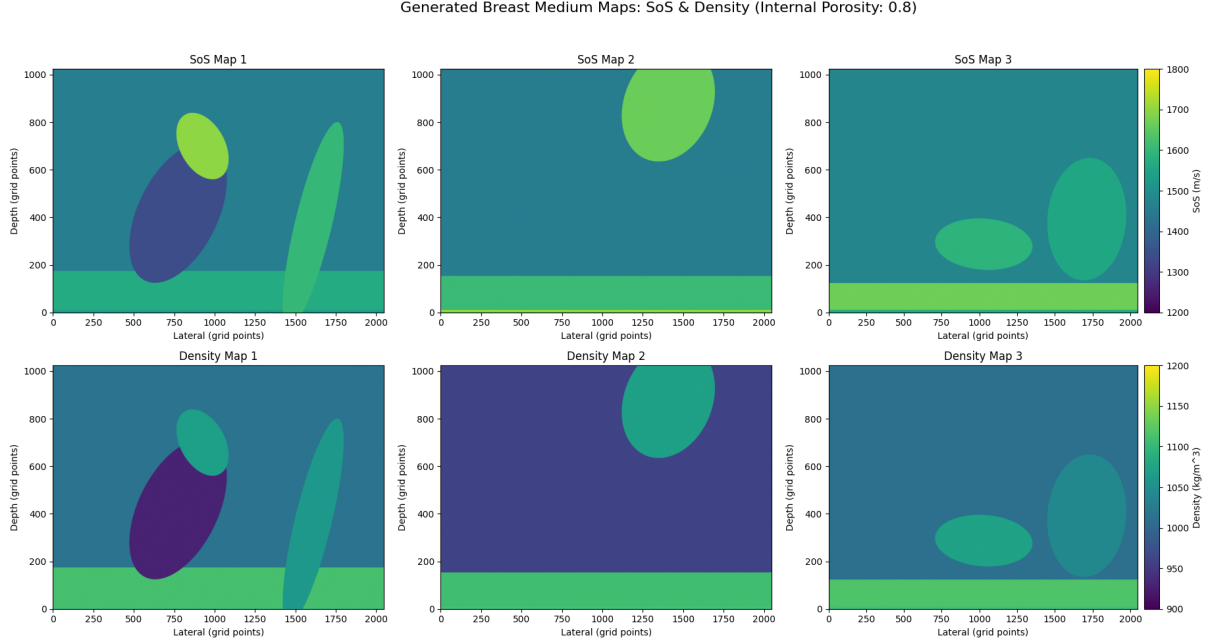


Figure 2: Generated Breast Medium Maps for SoS and Density. These maps serve as the computational phantom, utilizing a 2048×1024 grid to represent various tissue distributions and inclusion geometries for ultrasound simulation.

2 Material Properties and Calculations

2.1 Speed of Sound (c)

The speed of sound in skin layers is highly dependent on the concentration of structural proteins. Following the methodology of Moran et al. (1995), a linear regression model relates the spatially averaged speed of sound (c) to the percentage area of histological components (x):

$$c = a + b \cdot x \quad (1)$$

where a is the intercept and b is the regression coefficient. For the epidermis, the predictor is the percentage of **Keratin** stain, while for the dermis, it is the percentage of **Collagen** stain.

2.1.1 Derivation Methodology and Error Propagation

The representative values for the breast skin layers were derived by applying the mean percentage area of fibrous protein content observed in the study's normal skin samples to the regression coefficients (a and b) provided in Table 4 of the reference.

- **Epidermis (Keratin Content):** Seven of the eight samples examined exhibited a mean keratin area ranging between 45% and 65% Moran et al. (1995).

- **Dermis (Collagen Content):** The samples showed a narrower distribution for collagen, ranging from approximately 58% to 64% by area Moran et al. (1995).

To account for the uncertainty in both the regression coefficients and the biological variability of the samples, the standard deviation (σ_c) was determined using the following error propagation formula:

$$\sigma_c = \sqrt{\sigma_a^2 + (x \cdot \sigma_b)^2 + (b \cdot \sigma_x)^2} \quad (2)$$

Based on these calculations, the consolidated acoustic velocities used for the simulation are:

- **Epidermis:** $c = 1633.7 \pm 101.8 \text{ m s}^{-1}$
- **Dermis:** $c = 1624.7 \pm 55.8 \text{ m s}^{-1}$

2.1.2 Comparison with Literature Data

The calculated values align closely with historical measurements across varying frequencies. Figure 3 illustrates the speed of sound measurements from this study (at approximately 26 °C) superimposed on historical data for human and animal skin tissues.

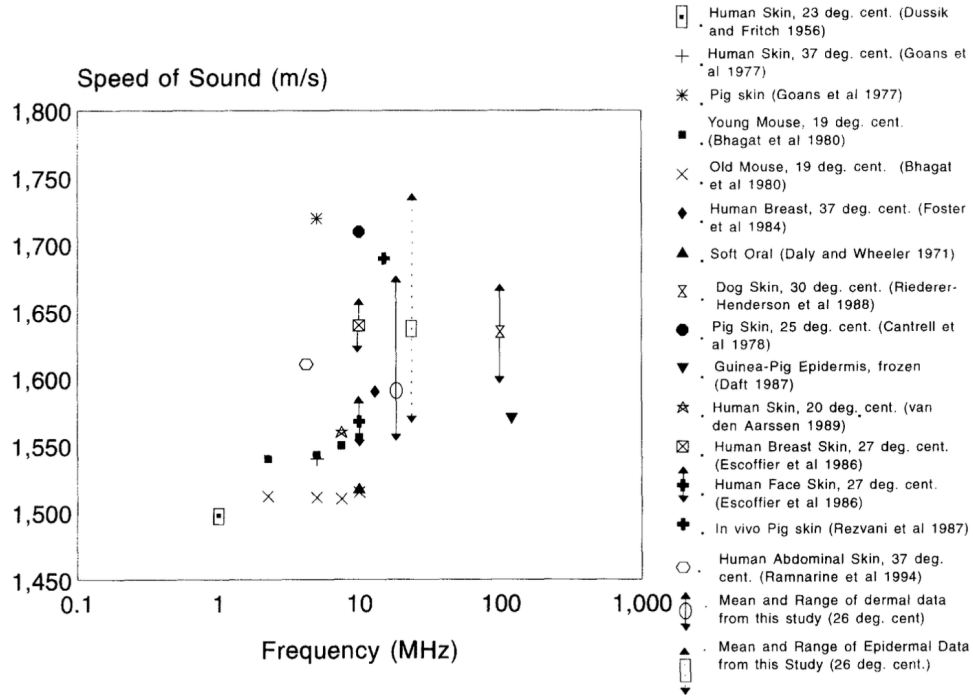


Figure 3: Speed of sound at various frequencies. Data from Moran et al. (1995) for dermal and epidermal layers are shown relative to previous literature findings (e.g., human breast, pig, and mouse skin).

- **Epidermis (Keratin):** $c = 1633.7 \pm 101.8 \text{ m s}^{-1}$
- **Dermis (Collagen):** $c = 1624.7 \pm 55.8 \text{ m s}^{-1}$

2.2 Layer Thickness (h)

Thickness data for the breast region are derived from regional histometric studies and anatomical surveys:

- **Epidermis Thickness:** $120.6 \pm 36.1 \text{ } \mu\text{m}$ (Measured using a new Ethical Histometric Technique).
- **Dermis Thickness:** $1337.9 \pm 332.3 \text{ } \mu\text{m}$ (Based on Korean adult population data).

2.3 Density (ρ)

Density is standardized based on the IT'IS database for biological tissues to maintain physical consistency in acoustic modeling:

- **Tissue Density:** $1109 \pm 13.89 \text{ kg m}^{-3}$ (IT'IS Database V5).

3 Consolidated Results Table

The following table summarizes the parameters for computational acoustics and B-mode beam-forming simulations.

Table 1: Averaged Skin Properties for the Breast Region

Layer	Thickness (h) [μm]	Speed of Sound (c) [m s^{-1}]	Density (ρ) [kg m^{-3}]
Epidermis	120.6 ± 36.1^a	1633.7 ± 101.8^b	1109 ± 13.89^c
Dermis	1337.9 ± 332.3^d	1624.7 ± 55.8^b	1109 ± 13.89^c

^a Oltulu et al. (Ethical Histometric Technique); ^b Moran et al. (1995);

^c IT'IS Database V5; ^d Hwang et al. (2002, Korean Adults).

Regarding the ultrasound coupling medium, **Propylene Glycol** exhibits a speed of sound ranging between **1467.805** and **1541.057** ms^{-1} (according to Thol et al., 2021) and a density between **1036** and **1040** kg m^{-3} (according to REMPEC MIDSIS).

4 References

1. Griffiths, C. E. M., Barker, J., Bleiker, T. O., Hussain, W., & Simpson, R. C. (Eds.). (2024). *Rook's Textbook of Dermatology* (10th ed.). Wiley-Blackwell. ISBN: 978-1-119-70928-2.
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7. REMPEC MIDSIS. Propylene Glycol Chemical Data. Available at: <https://midsis.rempec.org/en/find-chemical/propylene-glycol>